STATISTICS 475: Applied Discrete Data Analysis
MIDTERM 2
Mar 28, 2023

INSTRUCTIONS: **DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO!**

- Answer all parts of all problems in the space provided.
- Place final answers in the blanks when they are provided.
- **Show work!** If you don't have a calculator, it is enough to write out the exact equation with the right numbers so that I could calculate the result myself.
- If you need more space (I don't think you should), indicate clearly where the problem continues.
- Use $\alpha = 0.05$ unless told otherwise

PROBLEM 1 (6 pts)
Refer to the Hair and Eye Color description and code.

1. **(1 pt)** What type of model did we save in `mod.fit`? What are these models called?

   Answer: *Loglinear model (Poisson regression is less specific)*

2. **(1 pt)** What is the baseline category for this analysis?

   Answer: *Hair: Black, Eye: Brown*

3. **(1 pt)** Write out the formula for estimating the mean count of students with blond hair and blue eyes (me!).

   Answer: $\exp(4.2195 - 2.2736 - 1.2238 + 3.8212)$

4. **(1 pt)** It is natural to ask whether hair color is independent of eye color. Use the data to answer this question and report the results below.
   The LR test for the interaction term answers this question. Test stat is 146.44, p-value is $< 2.2 \times 10^{-16}$. *We reject the null hypothesis of independence and conclude that hair color and eye color are associated with each other.*

5. **(1 pt)** Which object contains odds ratios between two rows and columns of the contingency table?

   (a) `emm`
   (b) `con1`
   (c) `con2`
   (d) `con3`
   (e) None of the above

6. **(1 pt BONUS)** For reasons I can't explain, the script I borrowed some of this code from had the line of code to create `coef.mat` at the end, but nothing following it. It seems to create a set of coefficients for manually estimating something from the model. If we used this in `mcprofile`, what population quantity would we be estimating?

   It is the odds ratio representing the ratio of odds between blond and brown hair for a person with blue eyes compared to a person with brown eyes. Or (Brown/blond) hair for (Brown/Blue) eyes. Or can reverse order of eyes and hair in the ratios: Blue/Brown eyes for Blond/Brown hair.

7. **(Bonus)** If I wanted LR confidence intervals for comparing the odds of brown vs. hazel eyes for someone with brown hair to someone with red hair, what coefficients would I need to give?
   (EBrown, HBrown)−(EHazel, HBrown)−(EBrown, HRed)+(EHazel, HRed)
   1: 0 2: 0 3: 0 4: 0 5: 0 6: 0 7: 0 8: 0

   9: 0 10: 0 11: −1 12: 1 13: 0 14: 0 15: 0 16: 0

PROBLEM 2 (5 pts)
"Putting the *cat* in *cat*egorical." Which distributional model is best for modeling data from the following situations, Binomial, Multinomial, or Poisson?

1. My orange cat likes to chew on boxes. He bites off pieces and leaves the pieces on the floor. He's weird. During one week, I gave him a different box to chew on each day. I counted the number of pieces he left behind each day. How should I model these counts?
   BINOMIAL MULTINOMIAL **POISSON**

2. My black cat has been sick for a while. When she uses the litter box, sometimes she misses the box and sometimes she doesn't. For the last week, I have followed her to the litter box each time she has used it. I now know how many times she missed and how many she didn't miss (good kitty!). How should I model these counts?
   BINOMIAL MULTINOMIAL POISSON

3. My two cats are indoor cats, so they are lazy and get little exercise. They like treats. I throw food treats across the floor and they run after them. I did this 20 times last night and know how many treats each cat got. Now they are fat as well as lazy. How should I model these counts?
   BINOMIAL MULTINOMIAL POISSON

4. My two children each have a cat, so there are 4 cats in our family. We often post photos of our cats to our family group chat (I would say it's on SnapCat, but you wouldn't think that's funny right now.) When I downloaded them all after a week, I counted how many photos there were of each cat. How should I model these counts?
   BINOMIAL **MULTINOMIAL** POISSON

5. Both of our cats like to climb on the kitchen table. We always take them down, but they get back up. They don't respect us. For a week, I counted the number of times I had to take a cat off of the table each day. How should I model these counts?
   BINOMIAL MULTINOMIAL **POISSON**

PROBLEM 3 (7 pts)
Recall the horseshoe crab data from the last test. I now use Poisson regression to model the actual number of satellites (males near a mating female) around each female in the study. Refer to the results from the *Midterm 2 2023 HorseshoeCrabs.R* script.

1. **(1 pt)** In `mod1`, what does the estimate for `(Intercept)` measure in regard to satellites?
   The intercept is the log mean number of satellites for a female whose weight is zero.

2. **(1 pt)** In `mod3`, what does the estimate for `Weight` measure in regard to satellites?
   The `Weight` parameter is the change in log-mean number of satellites as the female's weight increases by 1 kg, holding width constant.

3. **(1 pt)** In `mod4`, what does the negative sign on the interaction term tell us about the relationship between the female's weight and the number of satellites she has?
   The relationship between weight and satellites changes at different widths. Specifically, when width gets larger, the effect of weight on the number of satellites decreases.

4. **(1 pt)** Consider the model in which the effects of the female's width and weight **do not** depend on the other measure. Report a 95% LR confidence interval for the ratio of mean numbers of satellites corresponding to a 1kg increase in weight, holding width constant at 26cm.
   From `exp(confint(mod3))`:
   CI= *(1.147, 2.135)*

5. **(1 pt)** Now consider the model in which the effects of the female's width and weight **do** depend on the other measure. Report a 95% LR confidence interval for the ratio of mean numbers of satellites corresponding to a 1kg increase in weight, holding width constant at 26 cm.
   From the `emmeans` results at `Width=26`:
   CI= *(1.55, 3.30)*

6. **(1 pt)** Consider the likelihood ratio tests for all effects in `mod4`. What do they suggest is the best model for us to use?

   (a) `mod1`

   (b) `mod3`

   (c) `mod4`

(d) None of the above

7. **(1 pt)** Considering all analyses, which of the following is the best conclusion?

   (a) Male satellites are more attracted to smaller females
   (b) Male satellites are more attracted to heavier females, regardless of her width
   (c) Male satellites are more attracted to wider females, regardless of her weight
   (d) **Male satellites are more attracted to heavier females with narrower shells**
   (e) Male satellites are more attracted to heavier females with wider shells
   (f) None of the above are accurate summaries.

PROBLEM 4 (3 pts)
This problem also relates to the horseshoe crab data, but focuses on analyses relating to the spine condition and shell colour. Refer to the results from the *Midterm 2 2023 HorseshoeCrabs.R* script.

1. **(2 pt)** A researchers wants to know whether the spine condition and colour of the crab are related. Report a test statistic and p-value to perform this test, and draw conclusions about the results.

   Test Stat= *34.74* p-value= 4.83×10^{-6}
   Conclusion:
   Spine condition is NOT independent of crab colour (or IS associated with or related to)

2. **(1 pt)** Regardless of whether this test is significant, which of the following best summarize any apparent pattern to the relationship between the spine condition and the shell colour?

   (a) **Crabs with a lighter shell are more likely to have better spine condition than darker crabs**
   (b) Crabs with a lighter shell are more likely to have worse spine condition than darker crabs
   (c) Crabs of `medium` shell colour are more likely to have better spine condition than lighter or darker crabs
   (d) Crabs of `medium` shell colour are more likely to have worse spine condition than lighter or darker crabs
   (e) There is no particular pattern to the relationship between spine condition and shell colour

PROBLEM 5 (3 pts)
Suppose that some measurement is believed to have a Binomial distribution, $Bin(4, 0.75)$—think of something like a 4-question quiz on Canadian trivia. As a result, there are specific probabilities on outcomes of 0, 1, 2, 3, or 4 successes. Let's call these probabilities $\pi_0, \pi_1, \pi_2\pi_3\pi_4$. You do not need to calculate them. We take 1000 samples of this measurement from a population and observe the following results:

Outcome	0	1	2	3	4
Frequency	5	45	215	425	310

The researchers want to test whether their measurement's distribution does indeed follow the probabilities that $Bin(4, 0.75)$ provides.

1. **(2 pts)** Write the formula for the test statistic, plugging in numbers and symbols as needed, but don't try computing it out.
   This is a Pearson Goodness of Fit test:

   $X^2 = \frac{(5-1000\pi_0)^2}{1000\pi_0} + \frac{(45-1000\pi_1)^2}{1000\pi_1} + \frac{(215-1000\pi_2)^2}{1000\pi_2} + \frac{(425-1000\pi_3)^2}{1000\pi_3} + \frac{(310-1000\pi_4)^2}{1000\pi_4}$
   It is acceptable to write this in summation terms as long as all symbols are clearly defined (e.g., $n_0 = 5, n_1 = 45, \ldots$)

2. **(1 pt)** Against what distribution would this statistic be compared to complete the test?
   χ^2_4.

PROBLEM 6 (Bonus)
Tom recently went on a rant about a problem that not many people are aware of. What was the problem?

1. Constructing confidence intervals by inverting hypothesis tests is not valid
2. **Inferences performed after variable selection are not valid**
3. Masks were effective against earlier strains of SARS-COV2, but not the more recent ones
4. Randomized clinical trials do not actually support causal conclusions
5. Tests for individual parameters in `glm()` are only approximate
6. Cats are superior to dogs